UNIVERSITI TEKNOLOGI MALAYSIA · FACULTY OF COMPUTING

PR3 FINAL DEMO · PR4 PRESENTATION

GreenStep

Small daily steps. Measurable climate impact.

Personal Carbon Footprint & Eco-Lifestyle Tracker — a cross-platform Vue 3 + PHP Slim 4 system

Vue 3 | Pinia | PHP Slim 4 | MySQL + PDO | JWT Auth | Capacitor

PROJECT	GROUP	LECTURER	FINAL DEMO
P3 · GreenStep	RM2Z · Section 02	Dr. Hishammuddin B. Asmuni	02 July 2026

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TEAM PROFILE & ROLES

Four leads, one integrated build

Zeng Liting

Frontend Lead · A24CS4073

- Vue 3 SPA scaffold, Router & Pinia
- 7 screen-by-screen views, mobile-first CSS
- Connected every page to the REST API

Mohammed Mohsen Alsakkaf

Backend & API Lead · A23CS4026

- PHP Slim 4 API · 9 controllers · 33 routes
- JWT middleware, CO2 + badge engines
- JSON envelopes, error handling, API docs

Rawan Mohamed Salih Magzob

Database & Security Lead · A24CS4066

- Authoritative MySQL schema & ER model
- Hardened PDO layer, prepared statements
- JWT/role policy, SECURITY threat model

Zineb Ennabili

DevOps, Mobile & Integration · A24CS4057

- Live deploy: Dockerized API (Render) + Netlify
- Admin RBAC: users · settings · tips · factors
- Auth hardening, integration & bug-fix sweep

GitHub Evidence

github.com/ennabilizineb/GreenStep

98 COMMITS · 4 CONTRIBUTORS · CLEAN HISTORY



Branching: work merged to `main`; release deliverables isolated on the **PR3** branch.

Existing carbon apps miss the middle

A substantial share of global emissions traces back to everyday choices in **transport, diet and energy**. Yet people rarely get immediate, quantifiable feedback — so habits never form.

Oversimplification

Failure mode 01

Static “green tips” (“turn off the lights”). No tracking, no calculation, no feedback loop — nothing to motivate a habit.

Overcomplication

Failure mode 02

Scientific calculators demand full life-cycle assessments — utility bills, exact distances, receipts. Cognitive overload → burnout → abandonment.

THE GAP

No app combines **low-friction daily logging** with **empirical, scientifically-grounded CO_2 calculations.**

“Individuals lack immediate, quantifiable feedback on the carbon implications of their daily actions.”

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OUR SOLUTION · GREENSTEP

Turn accountability into a 60-second daily habit

Log transport, meals, electricity and recycling with single-tap inputs. A PHP Slim 4 engine computes the footprint server-side against database-driven DEFRA / IPCC emission factors, then drives streaks, badges and community challenges.

OBJECTIVE 1

Multi-Role Secure Access

RBAC for User · Leader · Admin, all behind stateless JWT + bcrypt.

OBJECTIVE 2

Server-Side Calc Engine

Slim 4 REST API computes CO_2 from DB-driven factors via PDO.

OBJECTIVE 3

Low-Friction Cross-Platform

Vue 3 SPA, mobile-first, wrapped as an Android APK with Capacitor.

OBJECTIVE 4

Interactive Visualization

Weekly & monthly trends rendered from live JSON dashboards.

OBJECTIVE 5

Gamification & Retention

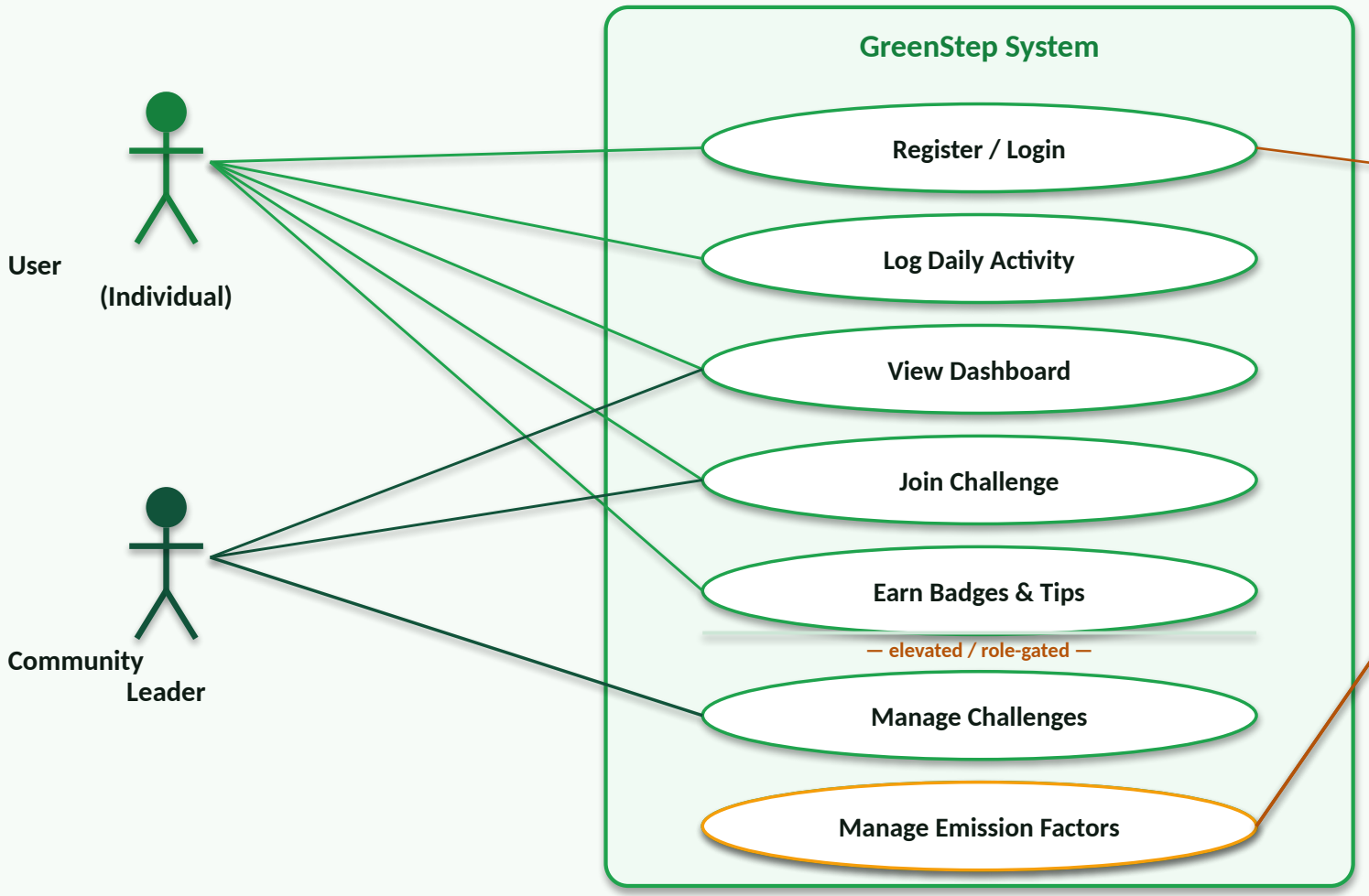
Automated streaks, JSON-criteria badges, collective challenges.

Scope delivered: Auth · Daily Log (full CRUD) · Dashboard · Challenges (CRUD) · Badges · Daily Tips · Admin factor & tip control — 3 roles, 10 tables, 20 endpoints.

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TARGET USERS & USE CASES

UML use-case diagram — three actors, one system



ROLE SCOPE (inherits downward)

U — User
Logs activities, views dashboard, earns badges, joins challenges.

L — Leader
All User rights + create / update / delete community challenges.

A — Admin
All rights + manage users, roles, tips, factors, badges & settings.

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TECHNOLOGY STACK

A decoupled, full-pipeline cross-platform system

PRESENTATION
Vue 3 SPA · Composition API

Vue Router 5 | Pinia 3 | Vite 8

MOBILE
Single codebase → native Android

Capacitor | Android APK | iOS-ready

APPLICATION / API
PHP 8.1 · Slim 4 RESTful micro-framework

7 Controllers | JWT Middleware | PDO · JSON

DATA
MySQL 8 relational store

10 tables | PDO prepared | FK cascade

DEVOPS / DELIVERY
Laragon → Docker → cloud

GitHub | Render | Netlify

WHY THIS STACK

Decoupled

Frontend is pure presentation; the API is a headless, stateless engine.

Authoritative

All CO_2 maths run server-side — clients never compute footprints.

Portable

One Vue codebase ships to web and Android via Capacitor.

Free-tier

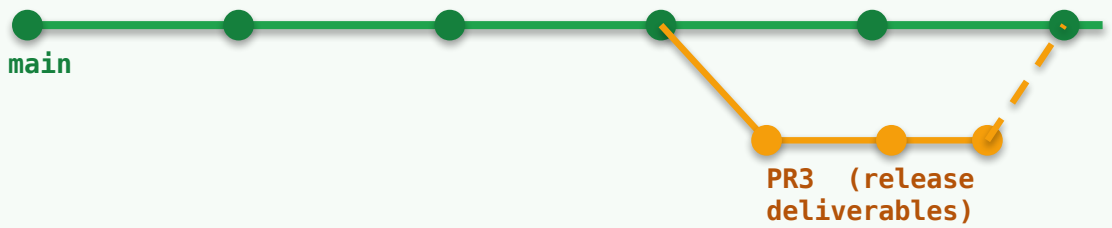
Render + Netlify + Cloud SQL = a public URL at zero hosting cost.

One repository, four contributors, clean history

github.com/ennabilizineb/GreenStep

Monorepo — Backend/ (Slim 4) + Frontend/ (Vue 3) + SQL, README & SECURITY docs.

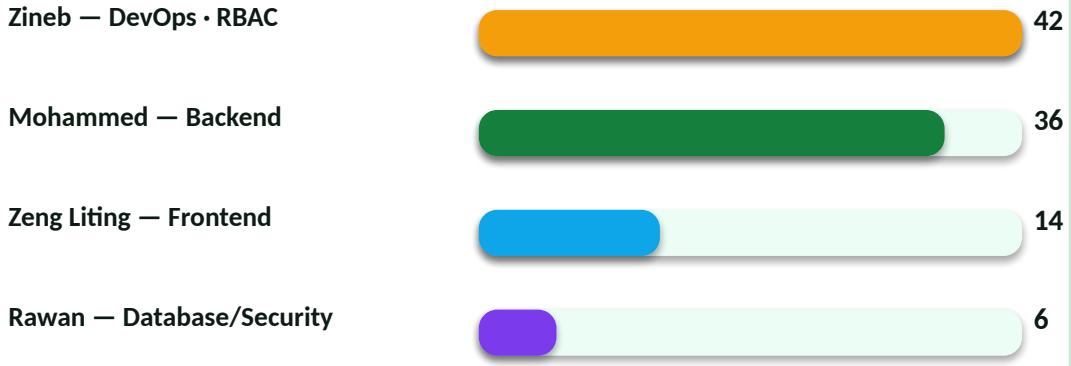
BRANCH STRATEGY



All members commit directly, merge into **`main`**; PR3 slides, README & Deployment docs are staged on a dedicated **PR3** branch.

CONTRIBUTION EVIDENCE

98 commits by verified author e-mail



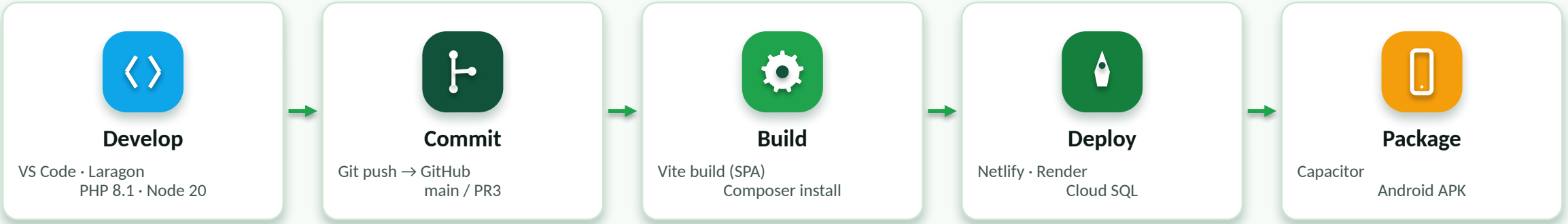
REPRESENTATIVE COMMITS

- **Backend:** Implement badge evaluation & collective challenge progress
- **DevOps:** Add Dockerfile · env-var API URL · admin RBAC controllers
- **Frontend:** Connect dashboard & challenges pages to backend API
- **Database:** update database schema and seed data · SECURITY.md threat model

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DEVOPS · CI/CD & DEPLOYMENT

From a Laragon laptop to a public cloud URL



Deployment Architecture

Frontend — Netlify / Vercel
Static Vue 3 build, auto-rebuild on push

Backend API — Render
PHP Slim 4 service over HTTPS

Database — Google Cloud SQL
Managed MySQL 8 via PDO

Mobile — Capacitor
Vue build wrapped as Android APK

TOOLCHAIN & STATUS

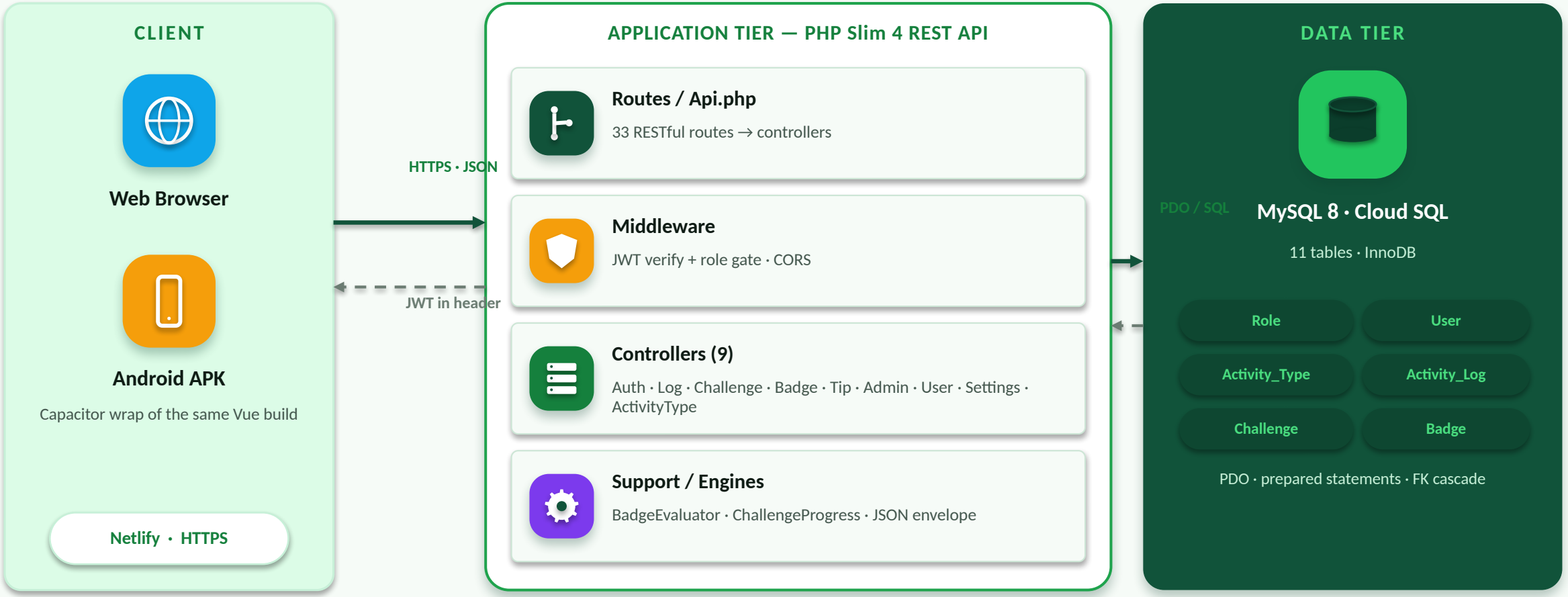
Laragon · Composer · npm / Vite · Postman · HeidiSQL · GitHub · VS Code · Capacitor

CI signal: Vue lint (oxlint + ESLint) and `composer validate` gate every build; Postman collection verifies all 20 endpoints before the demo.

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SYSTEM ARCHITECTURE

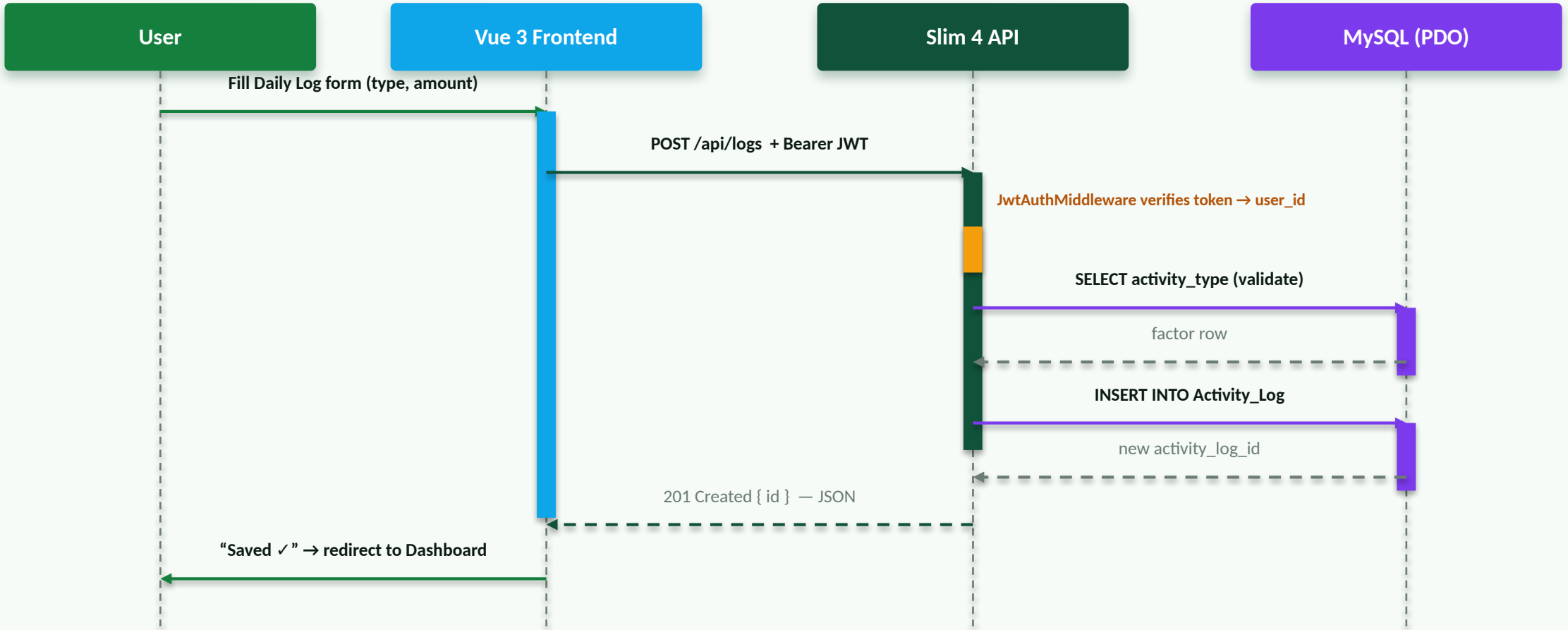
Three decoupled tiers communicating over REST + JWT



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SEQUENCE DIAGRAM

UML — logging a daily activity, end to end

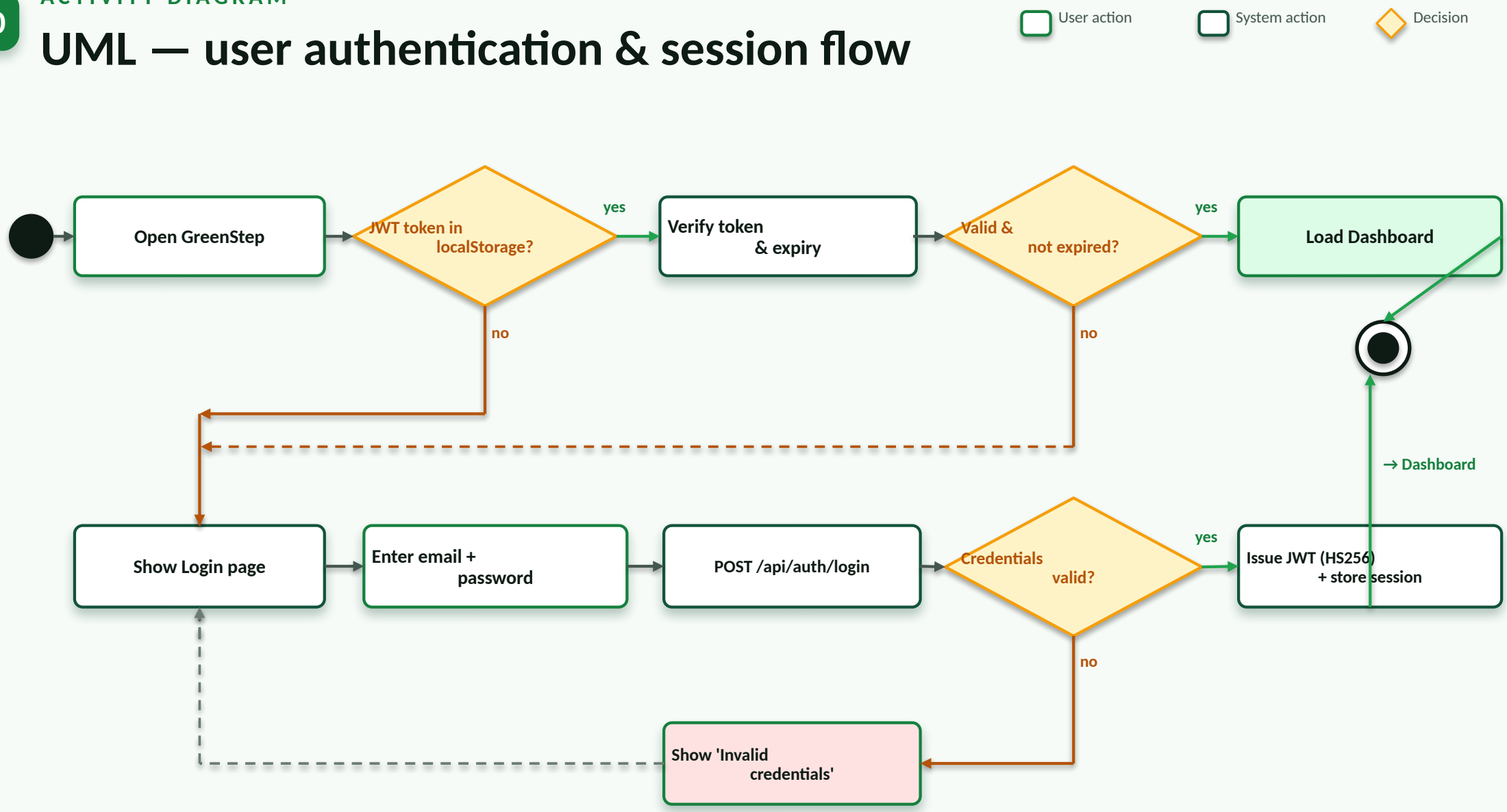


CO_2 is computed server-side later on `GET /api/dashboard` as amount × kg_co2_per_unit — the client never calculates the footprint.

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ACTIVITY DIAGRAM

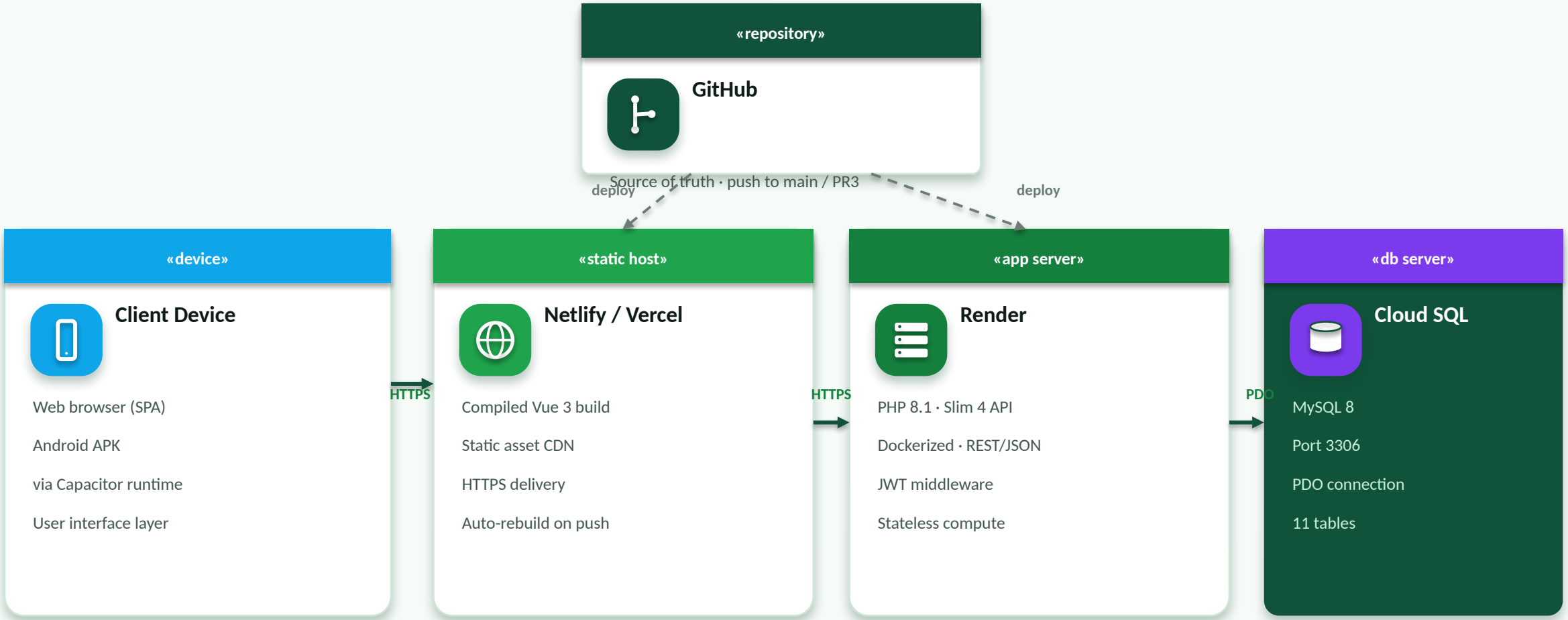
UML — user authentication & session flow



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DEPLOYMENT DIAGRAM

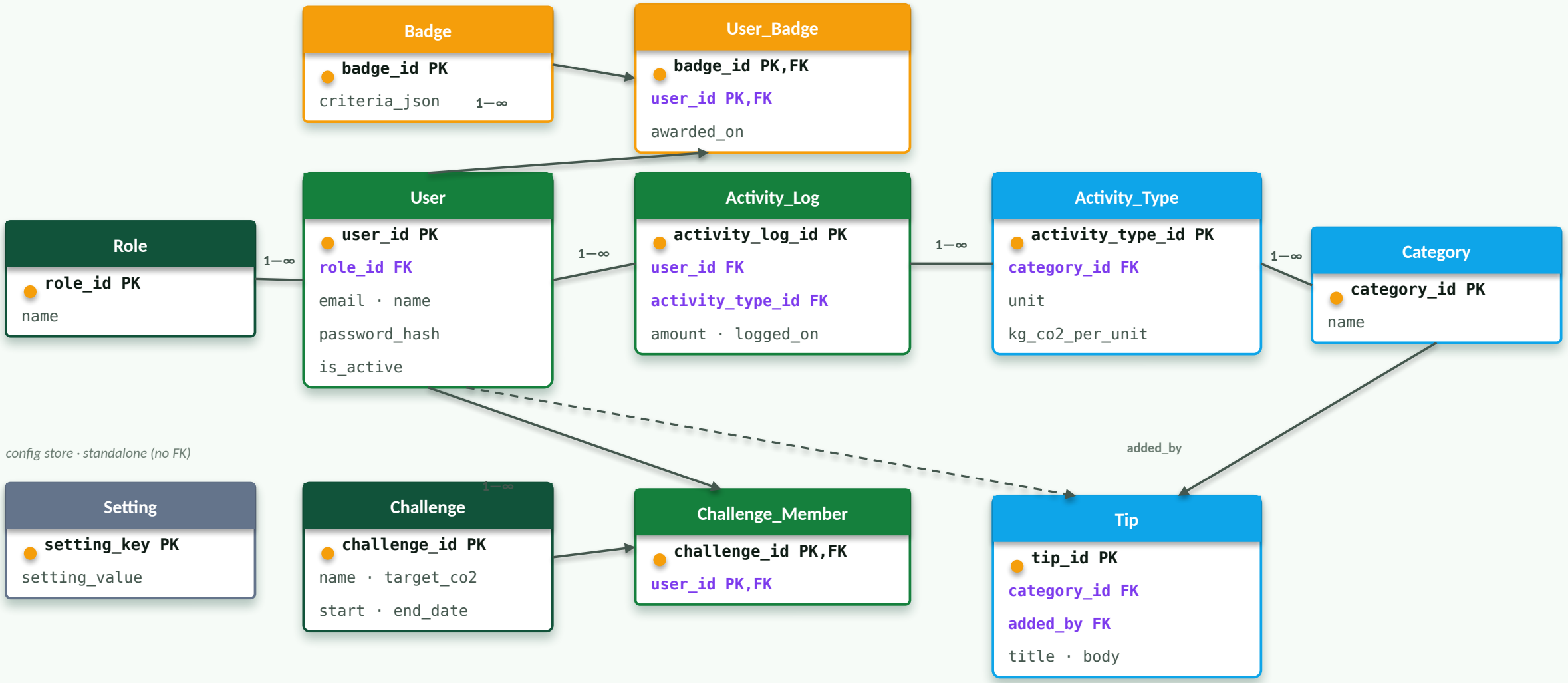
Nodes, hosting environments & network protocols



Free-tier public deployment: the browser/APK loads the Vue SPA from Netlify, which calls the Slim API on Render over HTTPS; the API reaches MySQL on Cloud SQL via PDO. GitHub drives every deploy.

DATABASE · ENTITY-RELATIONSHIP DIAGRAM

11 normalized tables + config store



PK Primary key FK Foreign key 1—∞ one-to-many All FKs use ON DELETE CASCADE / RESTRICT for referential integrity.

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BACKEND ENGINE · REST API & SECURITY

33 endpoints · 9 controllers · stateless JWT

API SURFACE — grouped by access level

4 — Public

`health · register · login · public settings`

11 — Authenticated User

`logs CRUD · dashboard · activity-types · tips · badges · challenges · join`

3 — Community Leader

`challenges: create · update · delete`

15 — Administrator

`users & roles · settings · factors · tips · badges · challenges · stats`

Inheritance: Leader ⊇ User ⊇ Public · Admin ⊇ all. Wrong role → **403**; missing/expired token → **401**.

Database.php

```php
// hardened PDO — structural SQLi defence
new PDO($dsn,$user,$pass, [
  PDO::ATTR_ERRMODE =>
      PDO::ERRMODE_EXCEPTION,
  PDO::ATTR_EMULATE_PREPARES => false,
]);  // params can never become SQL
```

AuthController.php — issueToken()

```php
$payload = [
  'sub'=>$id, 'role'=>$role,
  'iat'=>$now, 'exp'=>$now+$ttl ];
JWT::encode($payload,$secret,'HS256');
```

- bcrypt password_hash()
- time-safe verify
- HS256 JWT · 1h TTL
- user-enumeration safe
- FK cascade integrity
- uniform JSON envelopes

Authoritative server-side maths, credible factors

$$CO_2\ (kg) = amount \times kg_co2_per_unit$$

Activity	Unit	kg CO_2
Car	km	0.170
Public Transport	km	0.044
Electricity	kWh	0.212
Red-Meat Meal	meal	2.500
Mixed Meal	meal	1.200
Vegetarian Meal	meal	0.500
Recycling	item	−0.100

Source: DEFRA UK GHG 2023 + IPCC AR6. Factors live in the DB — admins update them via API, never in code. Negative = net carbon saving.

Badges

JSON-criteria badges (total_logs · streak_days · category_logs), auto-awarded; admins create & delete them.

Streaks

Consecutive-day algorithm walks back from today over distinct log dates.

Challenges

Collective baseline-vs-during window: Σ saved_kg vs target, ranked leaderboard.

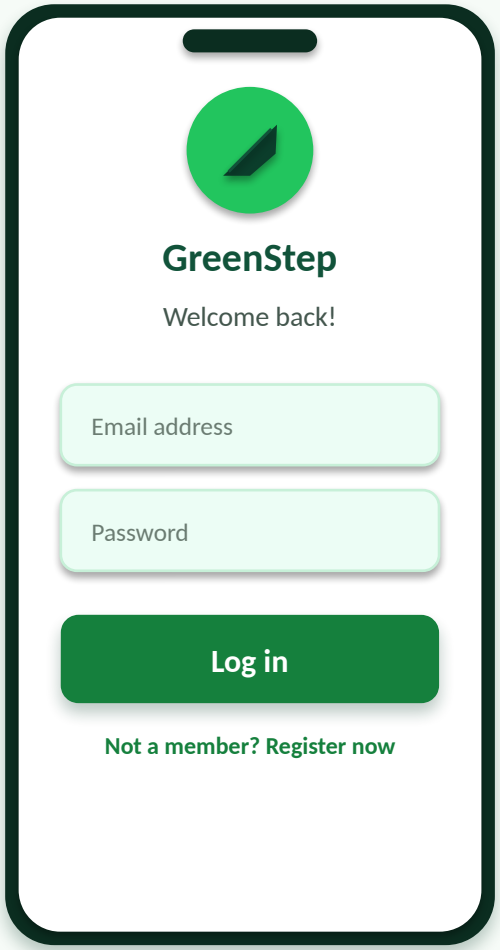
gamification rule

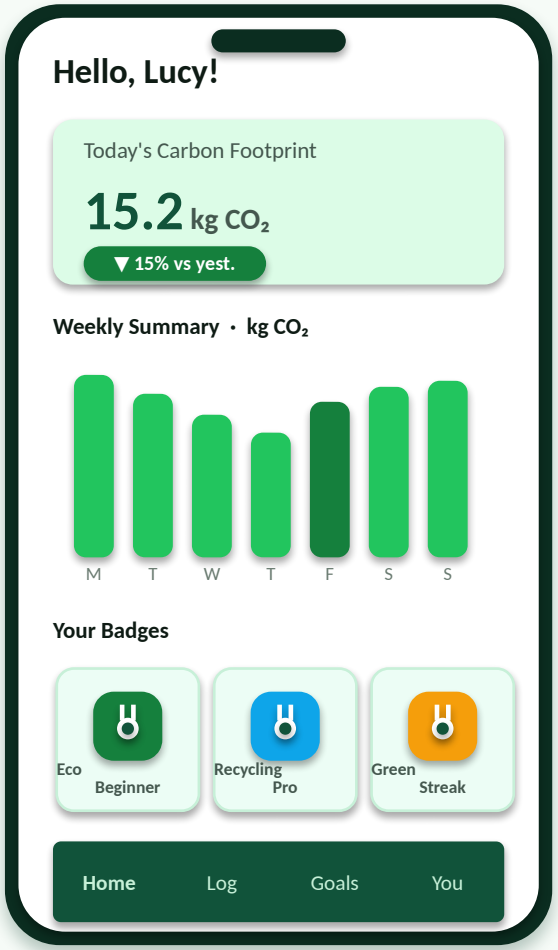
```
// Badge.criteria_json — evaluated server-side
{ "type": "streak_days", "threshold": 7 }
→ 'Green Streak' unlocks at 7 consecutive days
```

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LIVE DEMO · STEP 1

Authentication → personal dashboard





What the panel will see

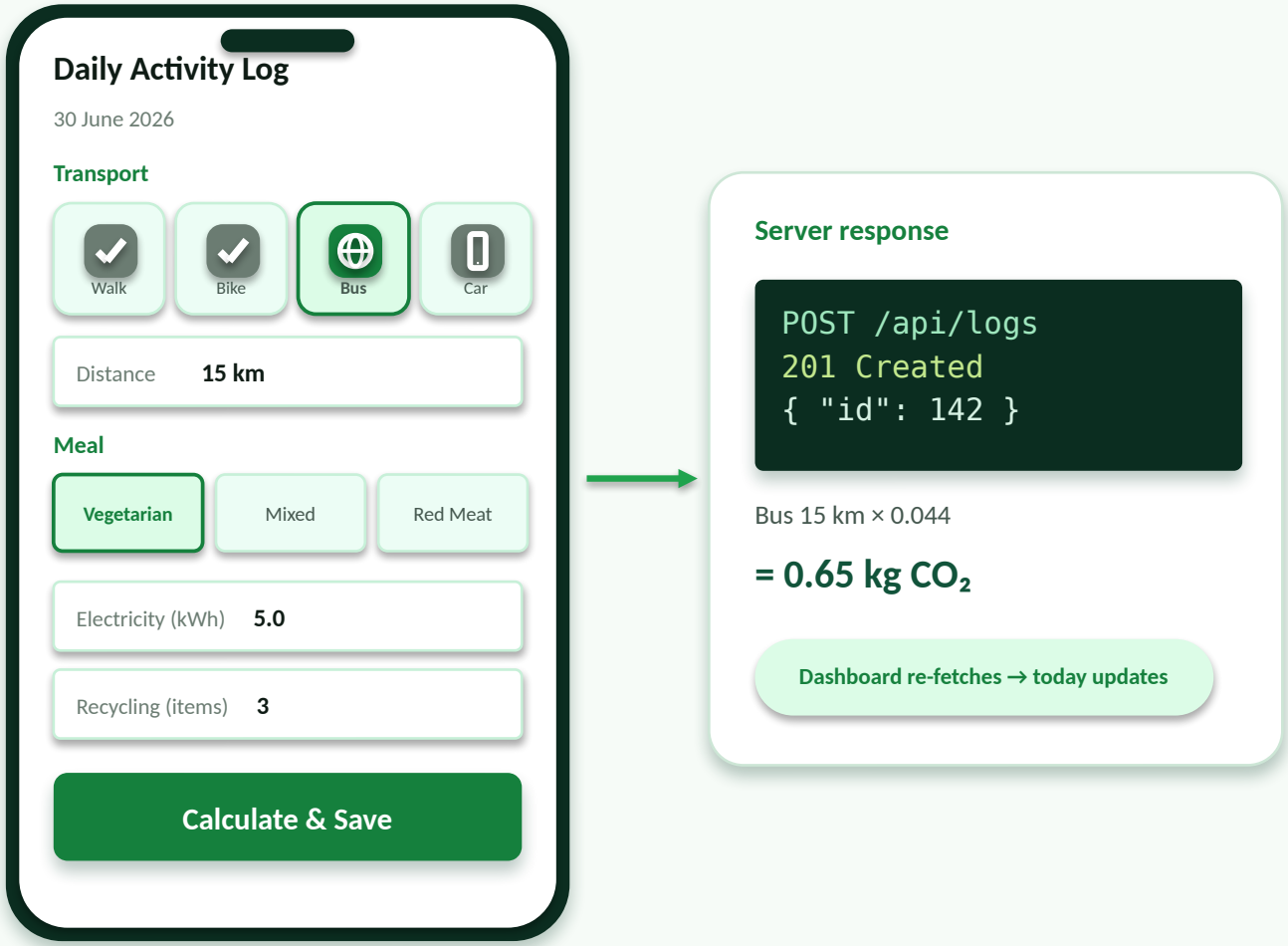
- Login posts credentials; the API verifies bcrypt + issues a JWT, stored client-side.
- Dashboard calls one endpoint; the server returns today's CO_2, the 7-day trend, streak & badges.
- Every number is computed live from seeded, realistic data — no hardcoded screens.

ENDPOINTS EXERCISED

```
POST /api/auth/login   →  200 { token, user }
GET  /api/dashboard    →  200 { today, week, badges }
```

LIVE DEMO · STEP 2

Log an activity → footprint updates instantly



Server response

```
POST /api/logs
201 Created
{ "id": 142 }
```

Bus 15 km × 0.044

= 0.65 kg CO_2

Dashboard re-fetches → today updates

Full CRUD, real calculation

- Activity types populate the form from the DB (GET /api/activity-types).
- Saving validates input, confirms the type exists, then inserts the log.
- The dashboard recomputes the footprint server-side on next fetch.

ENDPOINTS EXERCISED

```
GET  /api/activity-types
POST /api/logs   →  201 { id }
GET  /api/dashboard (recompute)
```

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LIVE DEMO · STEP 3

Challenges, badges & the admin control panel

Eco-Commuter Sprint

Public transit / cycling / walking for 7 days.

58%

3 days left

15 participants · target 15.5 kg saved

View Details

Badges — earned & locked

Eco Beginner

Green Streak

Recycling Pro

Carbon Cutter

Eco Warrior

ADMIN CONTROL PANEL

1,042
Total users

6,241
Activity logs

11
Challenges

31
Eco-tips

Manage: Users·Roles·Tips·Factors·Challenges·Badges·Settings

Collective progress

Challenge cards show member_count, days_left and a baseline-vs-during % computed across all members.

Automated rewards

Badges unlock from JSON criteria the moment thresholds are crossed — no manual awarding.

Credible & governed

Admins manage users & roles, tips, factors, badges and platform settings — role-gated.

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ENTREPRENEURIAL PITCH

From course project to commercial product · CLO3

CLO3

REVENUE MODEL 1

B2B Corporate SaaS

- Companies & universities chase ESG targets and staff engagement.
- Paid tier: private, closed-loop sustainability challenges for employees.
- Corporate dashboard with anonymized carbon analytics for reporting.

REVENUE MODEL 2

Affiliate Brand Partnerships

- Consumer app stays free; streaks & wins earn redeemable points.
- Points → discount codes from vetted eco-friendly brand partners.
- Revenue from partnership placement fees + affiliate commissions.

WHY USERS WILL TRUST THE NUMBERS

Institutional baselines

Factors sourced from DEFRA UK & IPCC AR6 — not invented.

Decoupled & governed

Factors live in the DB, admin-controlled; ready to pull live open-data registries.

Transparent methodology

Every figure shows its formula; optional camera proof for eco-actions.

Repositories, live access & test credentials

GitHub Repository

Monorepo · main + PR3 branch · Frontend / + Backend /

Full Project	`github.com/ennabilizineb/GreenStep`
Frontend	`github.com/ennabilizineb/GreenStep/tree/main/Frontend`
Backend	`github.com/ennabilizineb/GreenStep/tree/main/Backend`

Live System Access

Deployed: Netlify (frontend) · Render — Docker (API)

Live Frontend	*< paste Netlify public URL here >*

Test Credentials

All 3 roles · login identifier = email

User role 1
email: `member@greenstep.com`
pass: `password`

Leader role 2
email: `leader@greenstep.com`
pass: `password`

Admin role 3
email: `admin2@greenstep.com`
pass: `password`

Note: accounts are provisioned by **`schema.sql / seed.sql`**; login only verifies the hash + active status, so these seeded values work on the live database.

CONCLUSION

A complete, integrated cross-platform system

11	33	3	98	4
Tables	Endpoints	Roles	Commits	Leads

WHAT WE BUILT

- Vue 3 + Slim 4 + MySQL + JWT, fully integrated
- Server-side CO_2 engine on DEFRA / IPCC factors
- Gamification: streaks, JSON badges, group challenges
- Live: Dockerized API (Render) + Netlify + admin RBAC

WHAT WE LEARNED

- Cross-platform delivery from one codebase (CLO1)
- Secure REST API design — JWT, bcrypt, PDO (CLO2)
- An entrepreneurial, deployable product mindset (CLO3)
- Real Git collaboration across four contributors

CLO1 · Cross-platform | CLO2 · Innovative solution | CLO3 · Entrepreneurial

Thank you

Team RM2Z — Questions welcome

github.com/ennabilizineb/GreenStep · SCSM2223 · Section 02 · 2025/2026

Zeng Liting · Mohammed Alsakkaf · Rawan Mohamed Salih · Zineb Ennabili